



SOLAR HEAT

The intensity of the Sun fluctuates day-to-day and its heat causes changes in atmospheric pressure.



AIR PRESSURE

Low atmospheric pressure causes storms and strong winds. Prolonged high pressure can cause drought.

EXCITING WEATHER PHENOMENA

Weather can produce some amazing phenomena and rare sights.

SPRITES, ELVES, AND JETS

Sprites and elves are dancing red lightning flashes in the sky. Jets are cones of blue light on thunderclouds.

BALL LIGHTNING

This glowing orb lasts for only seconds. It may be caused when elements in the soil vaporize and react with oxygen in the air.

KATABATIC WINDS

These winds occur at night on mountain slopes. Dense, cold air is pushed down the slope by gravity.

NONAQUEOUS RAIN

Spiders, frogs, and even jellyfish can be whipped up in strong winds and then fall as rain.

ST ELMO'S FIRE

This electric spark is like the glow in a plasma ball but it occurs naturally on things like masts and lampposts during thunderstorms.

GIANT HAILSTONE

This whopper fell in Vivian, South Dakota, USA, during a July storm in 2010. Hailstones can gather ice layers as winds in storm clouds whip them upwards again and again.

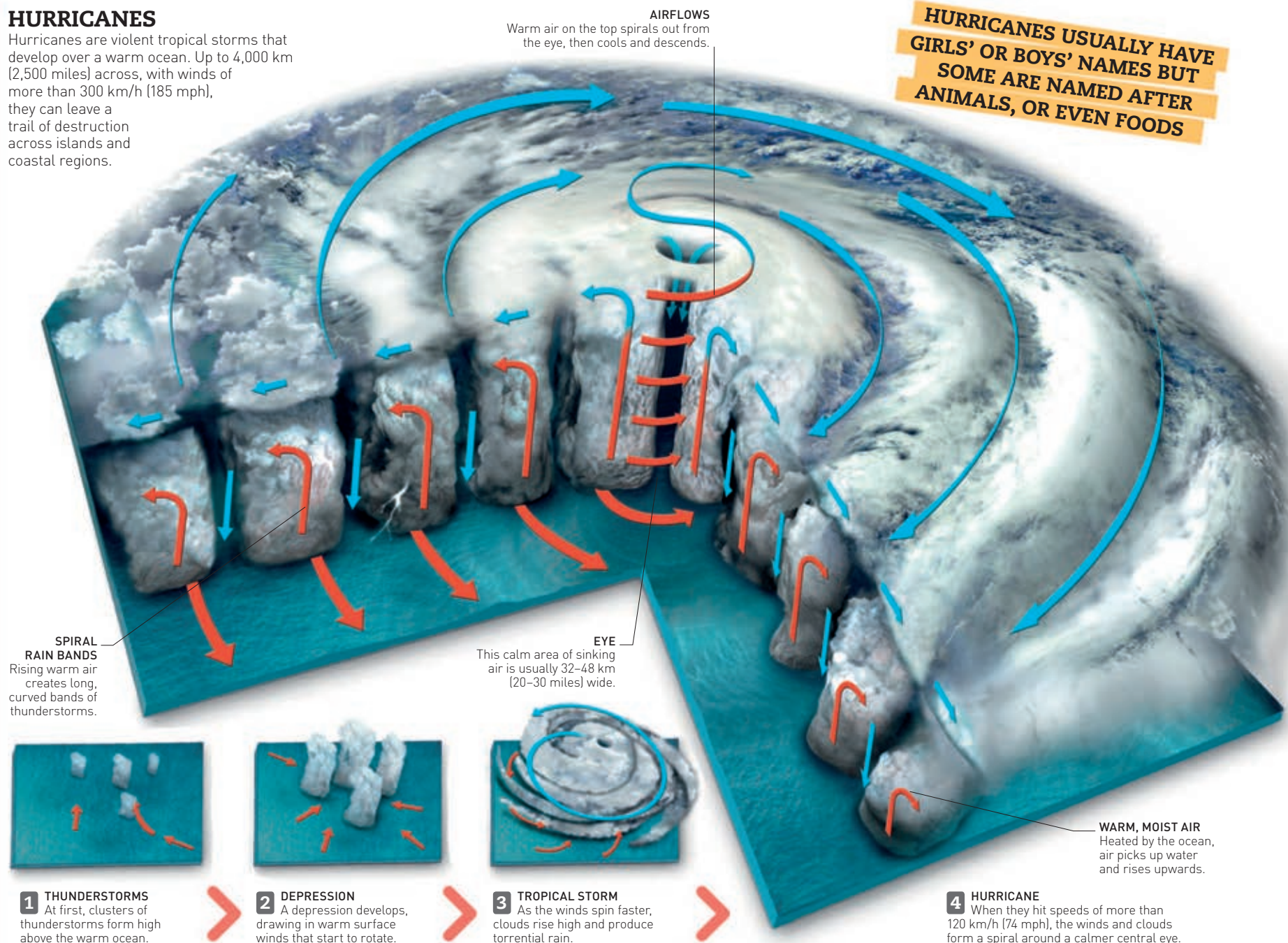


RECORD HAILSTONE

This hailstone weighed 1 kg (2.2 lb) and was 20 cm (8 in) across – three times the size of a tennis ball.

HURRICANES

Hurricanes are violent tropical storms that develop over a warm ocean. Up to 4,000 km (2,500 miles) across, with winds of more than 300 km/h (185 mph), they can leave a trail of destruction across islands and coastal regions.



HURRICANES USUALLY HAVE GIRLS' OR BOYS' NAMES BUT SOME ARE NAMED AFTER ANIMALS, OR EVEN FOODS

TORNADOES

These spinning columns of air reach from the clouds down to the ground. They are shaped like a funnel with a core of air that can spin at anything up to 480 km/h (300 mph). Nicknamed "twisters" in the USA, they form during summer storms and can destroy crops and buildings.



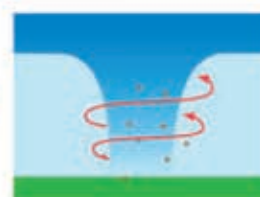
1 STEADY SPIN

As warm air rises from the ground it starts to spin. The base of a cloud forms a funnel.



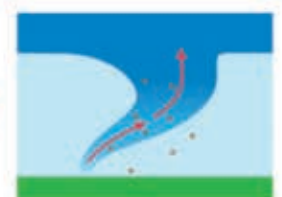
2 FUNNEL GROWS

When the funnel reaches the ground it draws in more hot air and begins to spin faster.



3 PEAK PROGRESS

At its peak, the column can be several kilometres wide and destroy everything in its path.



4 COLUMN DIES

After a while the column spins more slowly. It narrows and is drawn up into the cloud.